



Illegal Parking in Bus Stops

Revenue generation via fines & kerb fees



improved On-Time Performance of buses



Passenger-safety exposure



A self-funding enforcement & monetization programme

THE PROBLEM

346k

Annual Estimated Illegal Parking
at 912 stops carrying 76% of peak arrivals

~1.5 %

Peak On-Time Performance Loss
lost network-wide (60s per obstruction scenario)

2,900 h

Annual Bus Delay
AVL data and an 11-stop field survey

THE FIX PAYS FOR ITSELF

AED 46M

Year-1 fine revenue
(AED 200/violation, 56% ticket-to-cash) + kerb fees
growing to 4.2M/yr

+98M

5-yr NPV @7%
Hybrid fixed 512 stops + 20 portable + kerb monetization

< 1 year

Payback period
Revenue declines only because deterrence works



CRITICAL SAFETY ISSUE

Passengers board and alight in live traffic lane (field photo)

CRITICAL SAFETY ISSUE

Passengers step off into moving traffic



346,000 / year

obstructed arrivals — each one a potential boarding or alighting in the live lane



The most vulnerable pay first

elderly passengers, wheelchair users, parents with children — level kerb access is exactly what the layby exists to provide



Zero-distance near-misses

the door opens into the same lane cars use to overtake the stopped bus



A liability the Agency carries

every peer city (NYC, London, Singapore) criminalised bus-stop parking on safety grounds before revenue grounds



Field photo, 28 Jul 2026 — layby blocked by a private car; the bus serves the stop from the carriageway and the passenger alights on the road, not the kerb.

Documented daily, across the network



Private car holds the layby — the bus loads in the traffic lane beside it



Taxi and van parked hard against the marked, signed bus stop (route F16)



Vans occupy the stop — the Al Quoz New Housing bus is forced to serve from the lane

Who blocks the stops

Taxis & limousines
labour/shuttle vans
private cars

60 cases

Average 3-hour morning
at just 11 stops

*All photos: PTA field observations,
Jul 2026.*

The evidence: Morning Peak (7 to 10 AM), 11 stops



60

Average blocked-layby cases
(07:00–10:00)

1 in 21

AM-peak bus arrivals met
an obstructed layby

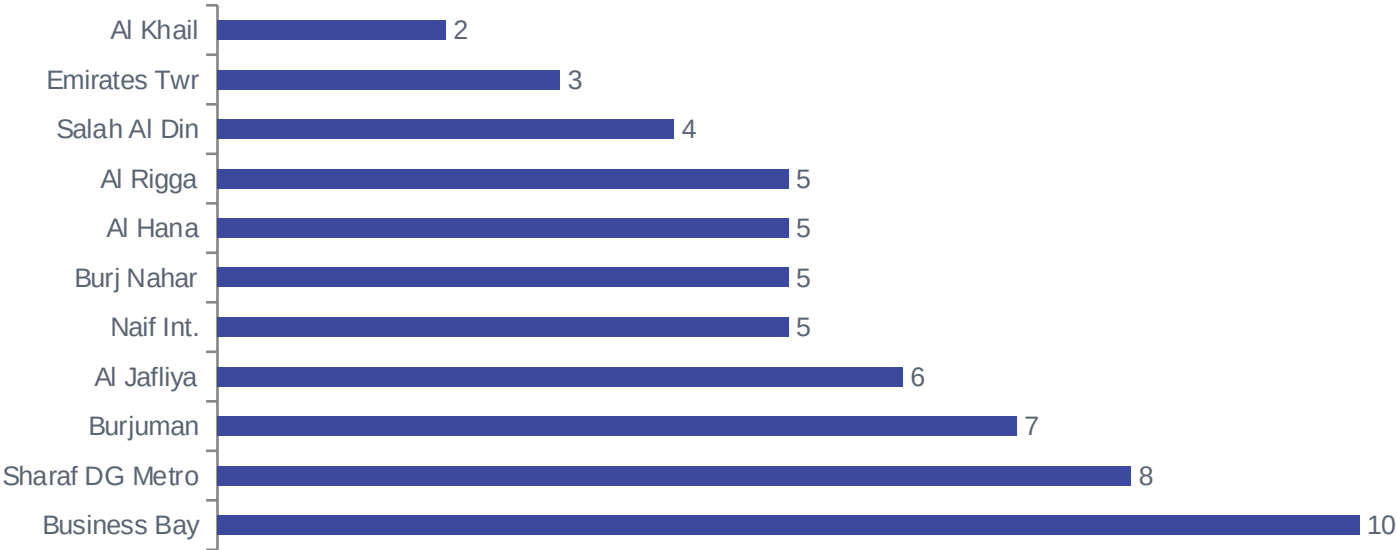
87.2%

baseline OTP that day
(38 routes, 72,679 stop events)

78.7%

OTP at Sharaf DG Metro —
worst stop, most blockages (8)

Blocked-layby cases by location (07:00–10:00, single day)



Cross-check

AVL data (22 Jul 2026) confirms these stops carry 1,254 bus arrivals in the AM peak alone — and that late running already concentrates in peak hours.

Network-wide: 912 stops carry the risk

Every stop in the GTFS network was scored on the surveyed hotspot profile: peak bus throughput plus metro-station or mall adjacency.

Tier	Stops	Definition	Share of peak arrivals	Annual blockages
High	112	Matches surveyed profile (metro/mall + heavy throughput) — survey-calibrated rate	24%	≈ 182,000
Medium	400	Next 400 by peak throughput — 50% of survey rate (assumption)	36%	≈ 134,000
Low	400	Following 400 — 25% of survey rate (assumption)	16%	≈ 30,000
Total	912	4 in 10 network stops	76%	≈ 346,000

76%

of all peak bus arrivals happen at these 912 stops

7.5

violations per stop per day at High-tier stops

Biggest non-surveyed exposures: Ibn Battuta, Union Metro, Mall of the Emirates, Gold Souq, Dubai Mall Metro, ADCB Metro.

What it costs the network every year

346,000

bus arrivals obstructed per year
(912 stops, both peaks)

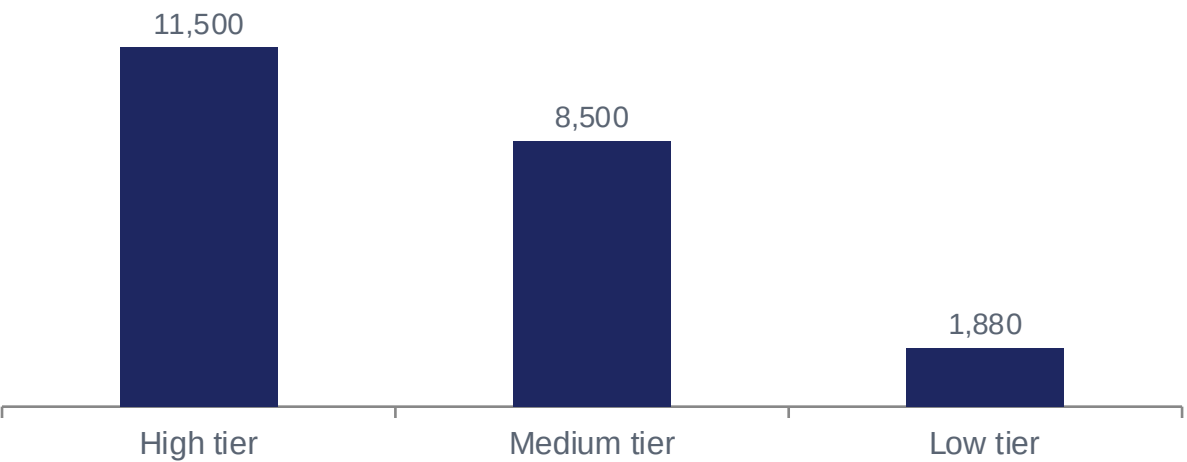
2,900 h

of direct bus delay per year
(at 30 s per obstruction)

−0.85 pp

peak OTP lost network-wide
(30–60 s scenarios: −0.35 to −0.85)

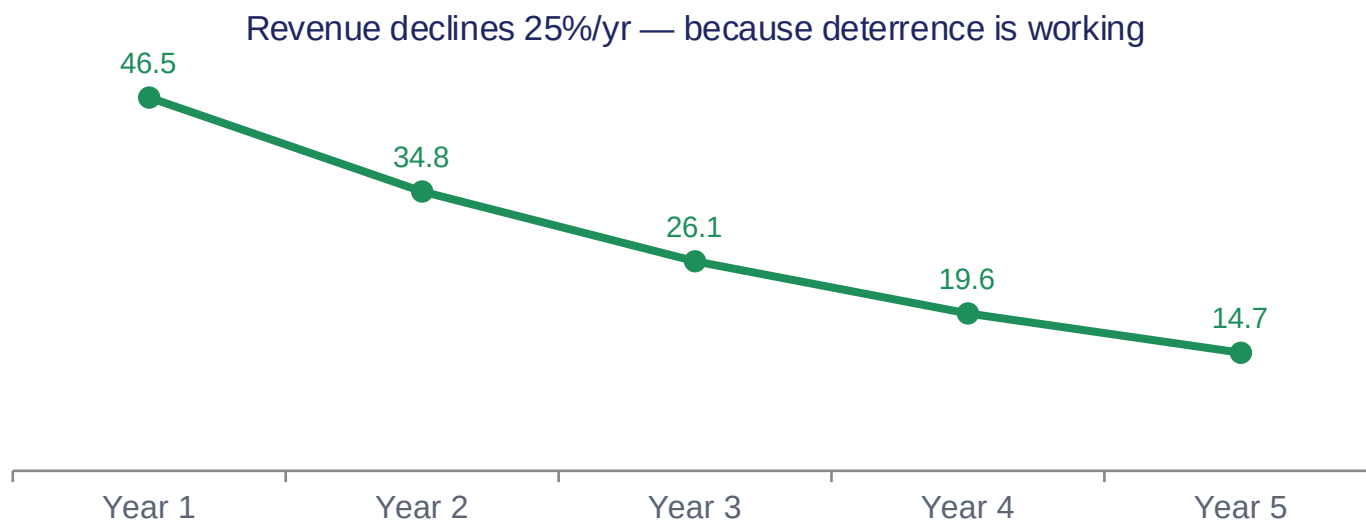
Extra late trips per year by tier (60 s per obstruction)



Why small delays matter

OTP counts a trip late beyond +5 minutes. The damage lands on trips already 4–5 minutes behind — 12.7% of peak stop events. A 30-second obstruction tips 1 in 38 of the trips it touches over the line; at 60 seconds, 1 in 16.

Enforcement pays: the violation-to-cash funnel



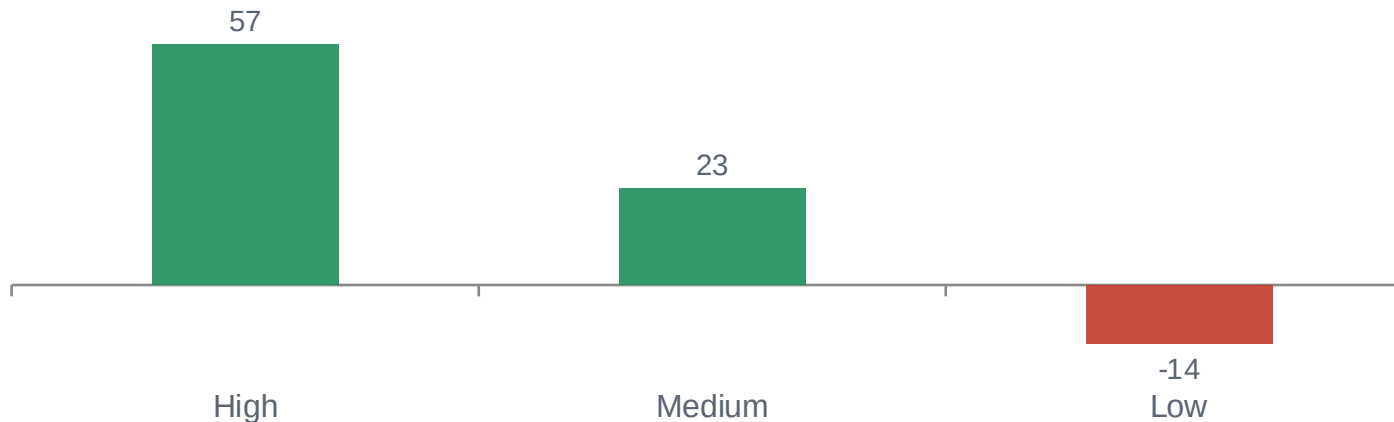
The honest pitch

Fines finance the system in years 1–3. The permanent returns are OTP recovery, safe boarding — and kerb fees that grow as fines shrink.

Option A — fixed cameras: invest where density pays

Tier	Capex	Revenue yr 1	5-yr NPV @7%	Payback
High — 112 stops	AED 4.5M	AED 24.4M	+56.6M	< 1 year
Medium — 400 stops	AED 16.0M	AED 18.0M	+22.6M	~1 year
Low — 400 stops	AED 16.0M	AED 4.0M	-13.8M	never

5-year NPV by tier (AED M)



Recommended

Stop at High + Medium.

512 stops · AED 20.5M capex
5-yr NPV +79.2M · payback < 1 yr

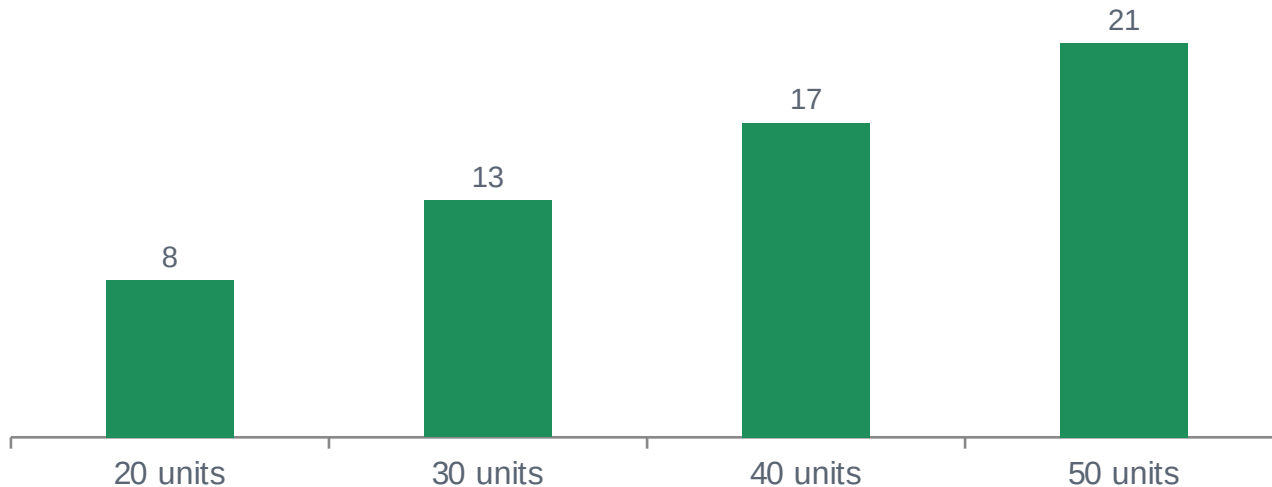
The Low tier's violation density is too thin to carry a fixed installation — leave it to the portable fleet.

Assumed: AED 40k/site capex, 5k/yr opex (benchmark; pre-RFP).

Option B — portable cameras: small capex, follows the problem



5-yr NPV by fleet size, monthly rotation at High-tier stops (AED M)



7.5

violations seen per
deployed camera per day

3.5x

daily revenue vs daily
all-in cost per unit

Deployed at High-tier stops; deterrence decay 15%/yr under monthly rotation.

Economics per unit: $\approx$ AED 830/day collected vs $\approx$ AED 240/day cost (AED 15k capex + 60k/yr crew, vehicle, redeployment). Every fleet size is NPV-positive — value scales with units, and rotation adds a "cameras could be anywhere" deterrence halo that fixed sites lack. Trade-off: only 2–5% of stop-time covered, so OTP recovery is slower than fixed coverage.

Monetize the kerb: charge operators for legal use



Enforcement creates a licensed product to sell: managed, off-peak layby access (max 3-min dwell, geofence-verified) — billed by the same ANPR cameras, so marginal field cost is ~zero.

AED 2.4M / yr

Annual layby-access permits

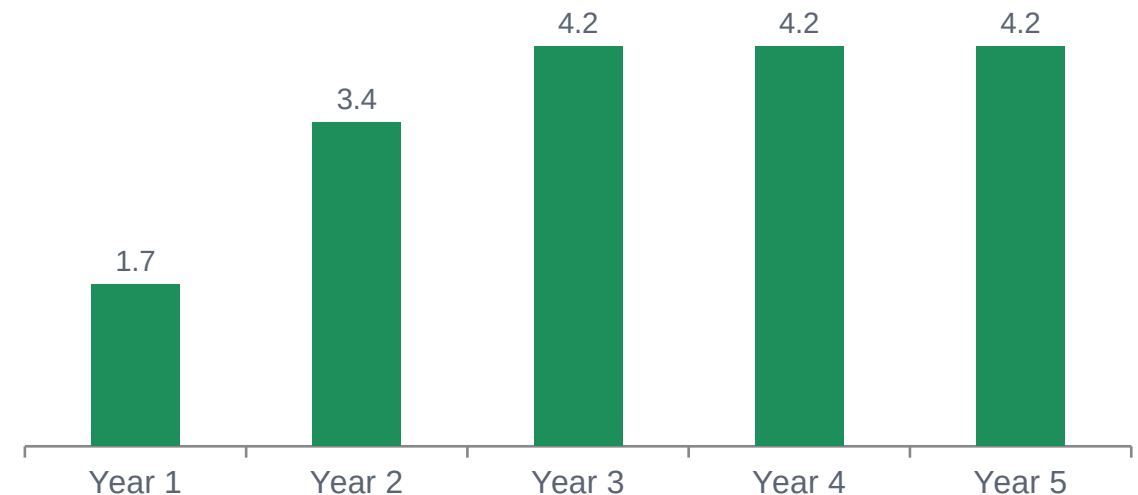
2,000 licensed shuttle & coach vehicles × AED 1,200/yr — off-peak windows only, never during bus peaks

AED 1.8M / yr

Pay-per-use kerb fee

1,000 paid dwells/day × AED 5, ANPR-billed — the airport kerb-fee model applied to bus laybys

Kerb revenue ramps up as fines ramp down (AED M)



+10.2M NPV · payback year 2 · and unlike fines, this line grows: it converts today's violators into paying customers with a legal option, and becomes the programme's permanent revenue as deterrence shrinks the fine curve. Setup: AED 2M platform + 0.5M/yr admin (assumptions, adjustable).

The 5-year money story — full programme

Fixed cameras at 512 High + Medium stops, 20 portable units, plus kerb monetization. All figures AED, undiscounted cash per year.



Recommendation: the hybrid programme



Phase 1 · Months 0–6

Fixed cameras at 112 High-tier stops + 20 portable units.
Run the validation survey that firms Medium/Low rates.
Capex ≈ AED 4.8M.

Phase 2 · Months 6–18

Extend fixed coverage to the 400 Medium-tier stops.
Launch layby-access permits for licensed operators.
Grow portable fleet to 30–50. Capex ≈ AED 18M — funded by Phase 1 revenue.

Phase 3 · Year 2+

Scale pay-per-use kerb fees — the permanent revenue line.
Portable rotation covers the Low tier and emerging hotspots.
Publish OTP recovery; review AED 200 vs AED 500 federal fine.

+98M

5-yr NPV, full programme (AED)

+0.85pp

peak OTP recoverable network-wide

< 1 yr

payback on every element deployed

Fines are the financing bridge; kerb fees are the steady state. Both decline/grow exactly as compliance improves — while OTP and safety gains compound.

Benchmarking: Every peer city has already moved



New York — MTA ACE

Bus-mounted cameras, escalating fines US\$50–250

~5% faster buses, ~20% fewer collisions, ~30% fewer repeat violations

London — TfL

Bus stop clearways, CCTV-enforced 24/7, PCN £160

Near-universal kerb access; compliance underpins the iBus OTP regime

Singapore — LTA

Camera-equipped buses since 2008, ~S\$150 fine

Sustained bus-lane and stop compliance

Sydney — TfNSW

Bus-zone offence ≈ A\$390

Deterrence through penalty level

Dubai today: manual patrols only. This study models a conservative AED 200 fine — the federal schedule already allows AED 500 for stopping in a bus stop, which would scale every revenue figure ×2.5 and turn even the Low tier positive.

Benchmarking: cities already charge for the kerb



Airports — SFO, LAX & most large US hubs

Kerb-access fees: US\$2.50–5 per pick-up

Taxis, ride-hail and shuttles billed automatically via geofence/AVI — the direct template for ANPR-billed layby dwell fees

Smart loading zones — Pittsburgh, LA, Columbus, Aspen

Camera-billed per-minute commercial kerb use

Registered operators pay per minute, violators get fined — enforcement and monetization on the same sensor

Central London — Westminster

Paid coach bays & permits

Coaches pay for timed use of high-demand kerb bays via permit/app

Chicago & US CBDs

Ride-hail pick-up zone surcharges

Designated TNC zones carry a per-trip fee for kerb access rights

The pattern: the same camera that fines illegal use bills legal use. Cities monetise scarce kerb instead of only policing it — RTA's laybys at 912 high-demand stops are exactly that asset.

Thank you